

# Duplicating Restricted Keys

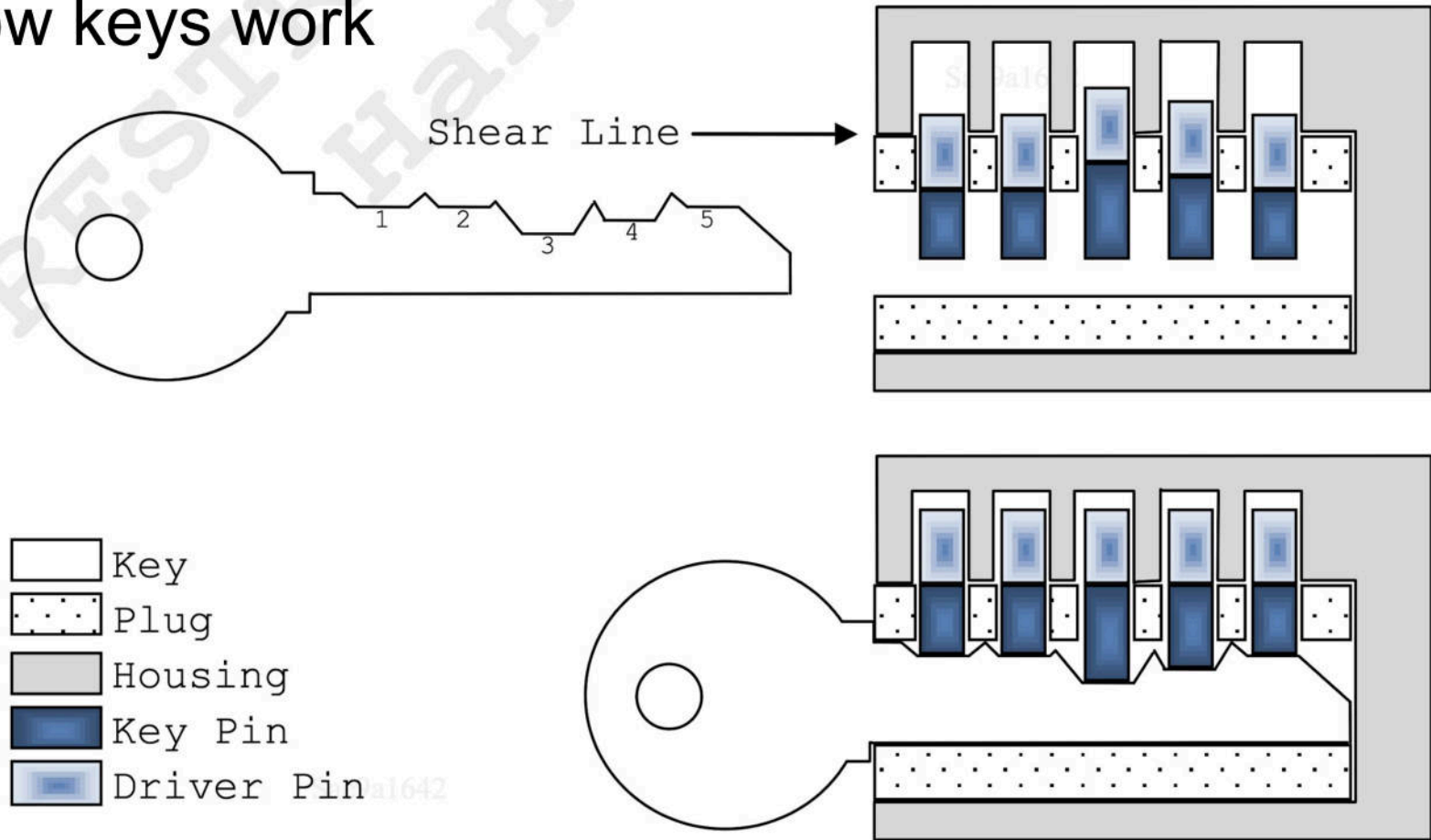
**Billy & Bobby Graydon**

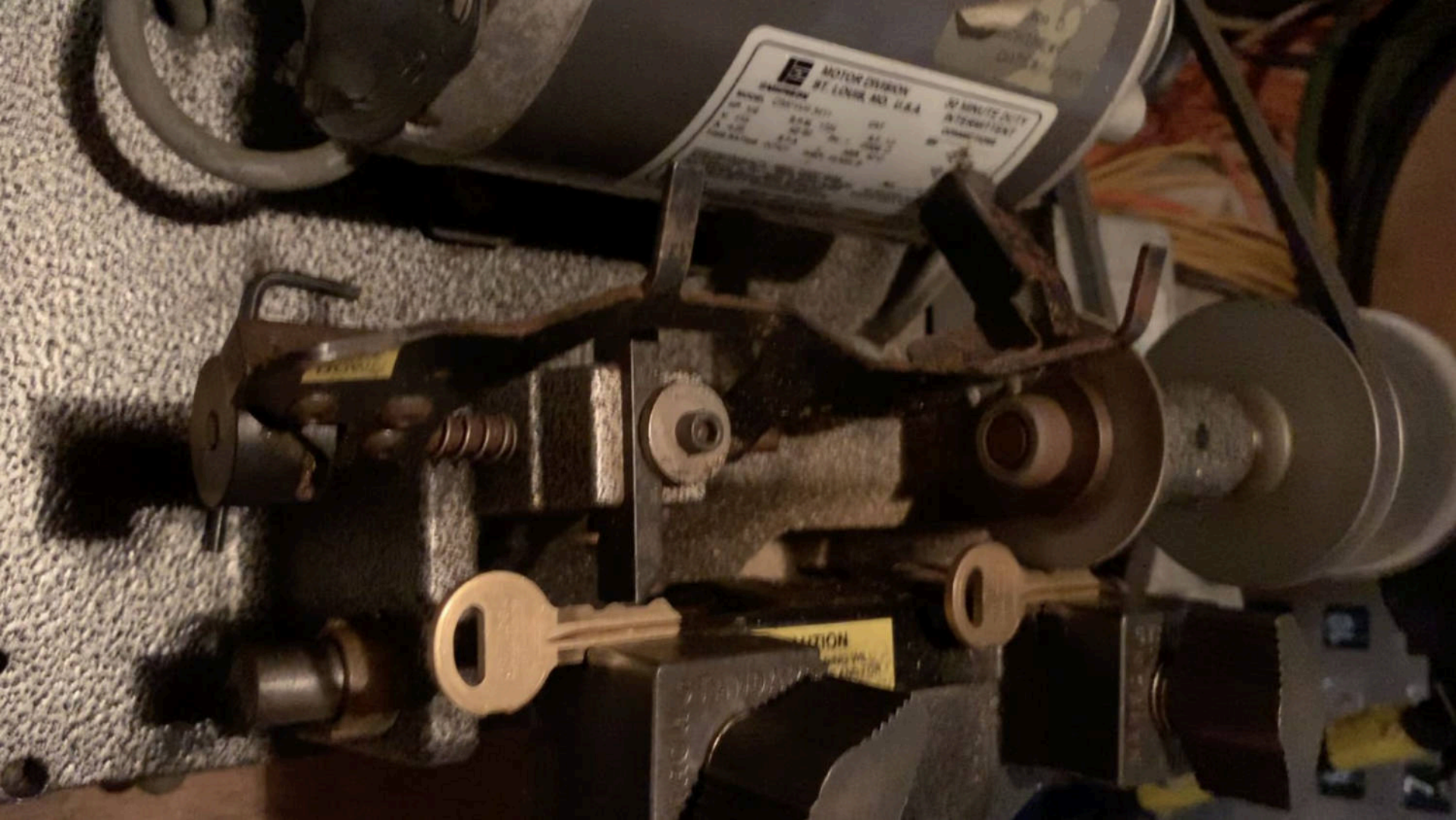


# Outline

1. Keyways
2. Other Stuff

# How keys work







# How keyways work



(1)



(2)



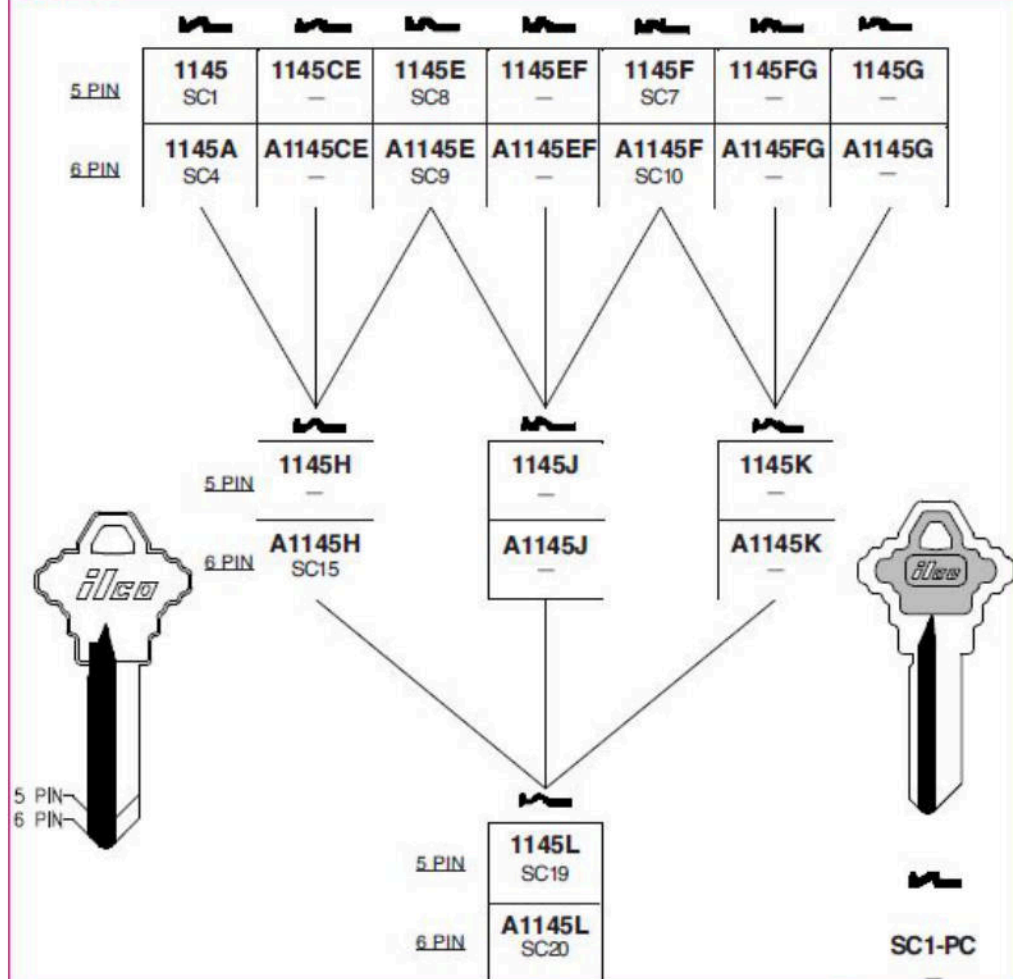
(3)



(4)



(5)





easy entrie

V9.0+  
Neusilber  
Version



OK



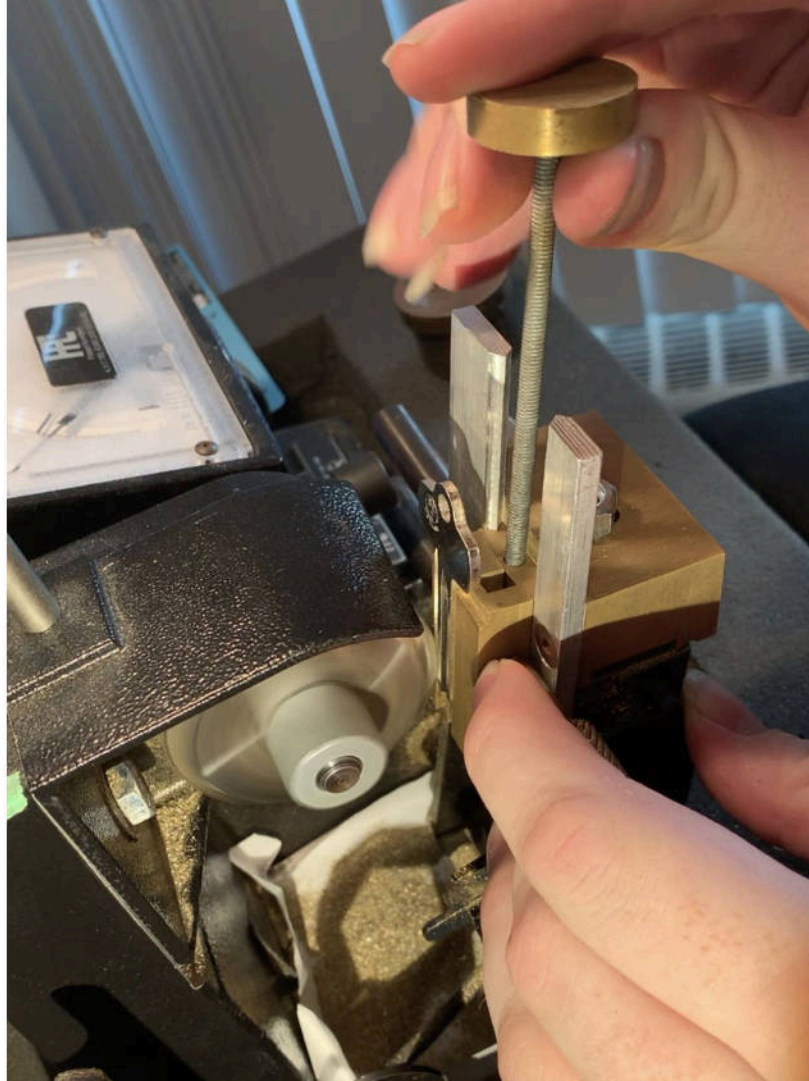
STOP

[www.KEYMAX.de](http://www.KEYMAX.de)  
Schlüssel ohne Wenn und Aber!









# The Old Approach to Keyway Research



# Our Approach to Keyway Research

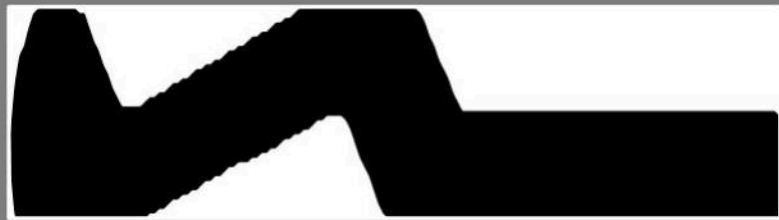
Medeco 16S 

Yale Y1 

Abloy AL1 

Kwikset 1176/KW1 

Schlage 1145/SC1 



draw Schlage 1145/SC1 

compare Schlage 1145/SC1 

Schlage 1145E/SC8 

compare

Sargent 1007RA 

reverse

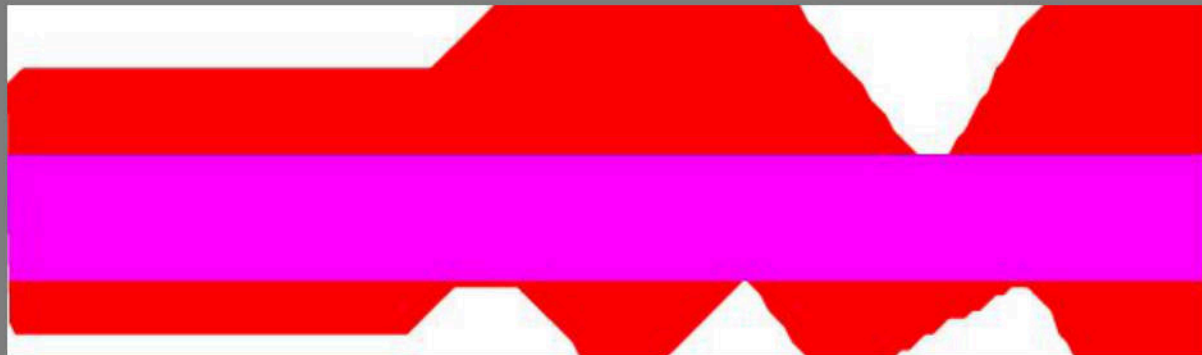
Sargent 1007LA 

compare

Medeco 9S 

barstock height:

, thick:





draw Schlage 1145/SC1 

draw Schlage 1145E/SC8 

draw Schlage 1145F/SC7 

draw intersect Schlage 1145/SC1   $\cap$  intersect Schlage 1145E/SC8   $\cap$  Schlage 1145F/SC7 

draw union Schlage 1145/SC1   $\cup$  union Schlage 1145E/SC8   $\cup$  Schlage 1145F/SC7 

C:



E:



F:



n:



U



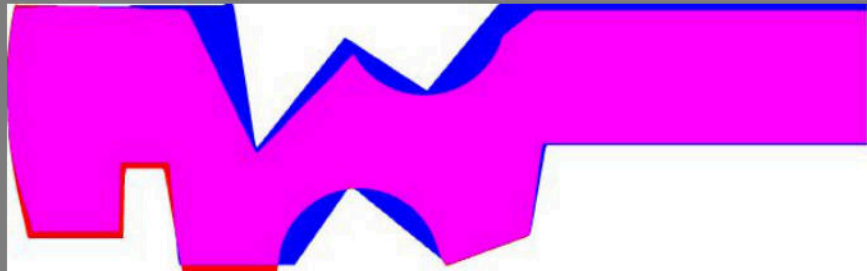
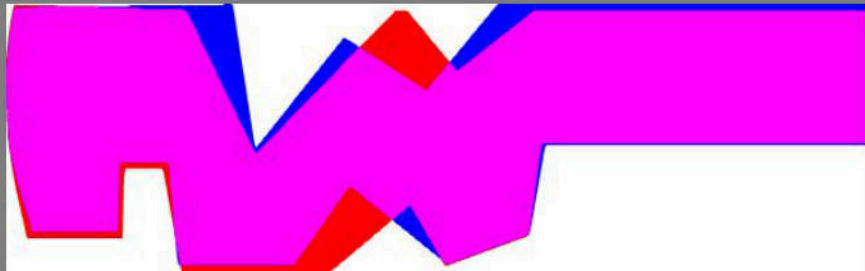
set Milled BEST L to BEST L 

compare Milled BEST L, BEST M 

set Milled BEST L to mill side A  of Milled BEST L with 62.5 dia ball cutter , depth: 35, position: 160

set Milled BEST L to mill side B  of Milled BEST L with 62.5 dia ball cutter , depth: 32, position: 135

compare Milled BEST L, BEST M 



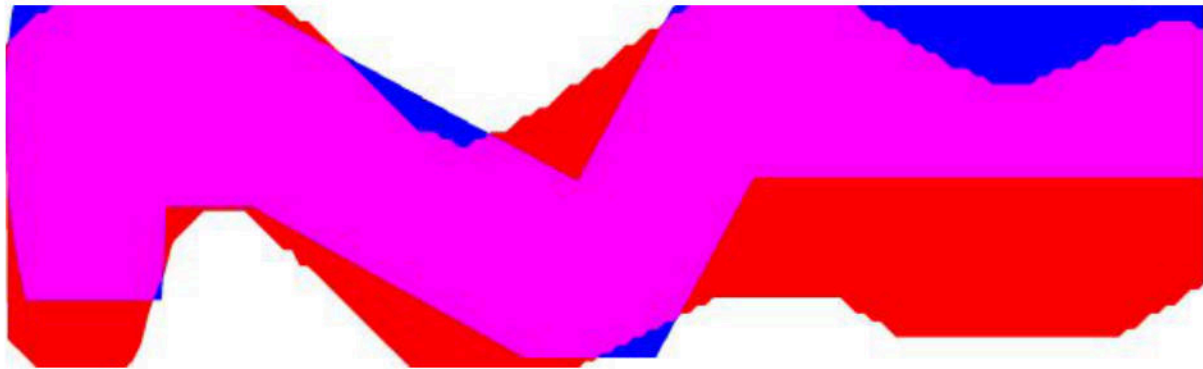


compare

Medeco 1515



BEST A









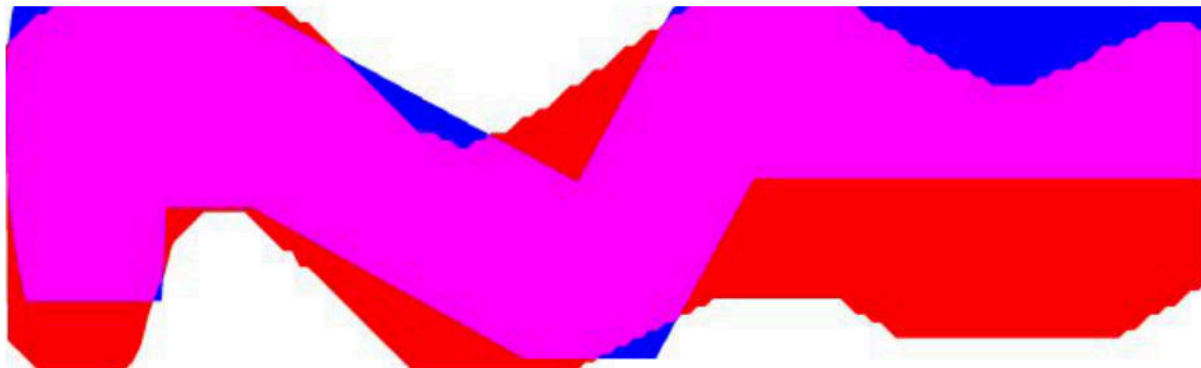
compare

Medeco 1515



,

BEST A













draw

Medeco 1515



draw

Schlage 1145E/SC8



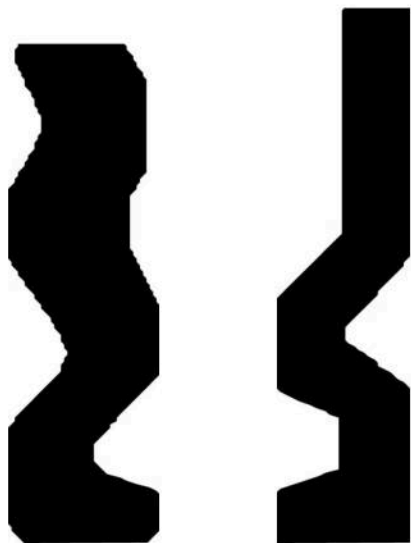
compare

reverse

Schlage 1145E/SC8



Medeco 1515



DII  
PAT. NO.  
3499302









medeco





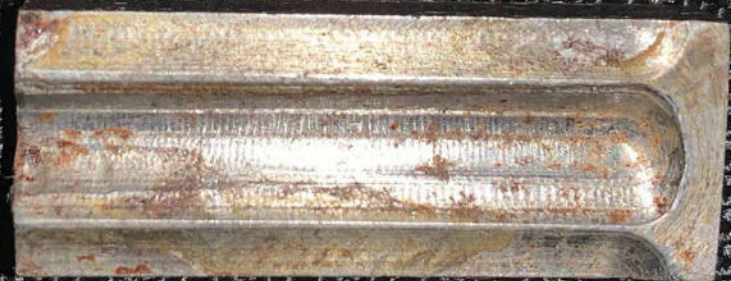








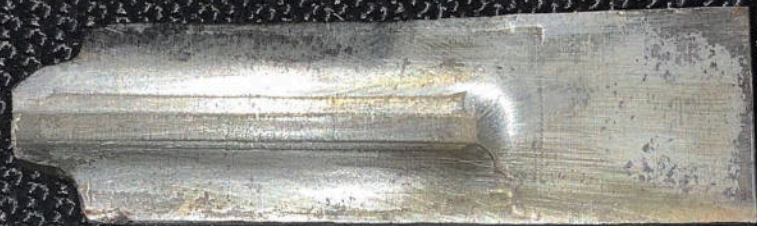












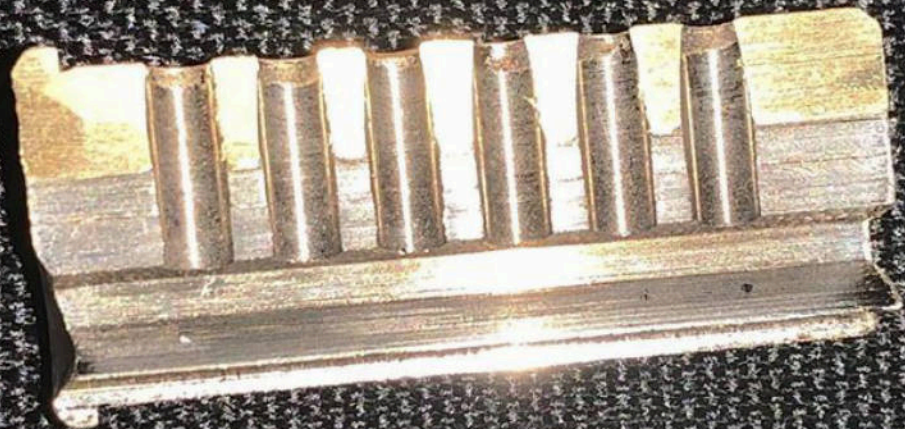




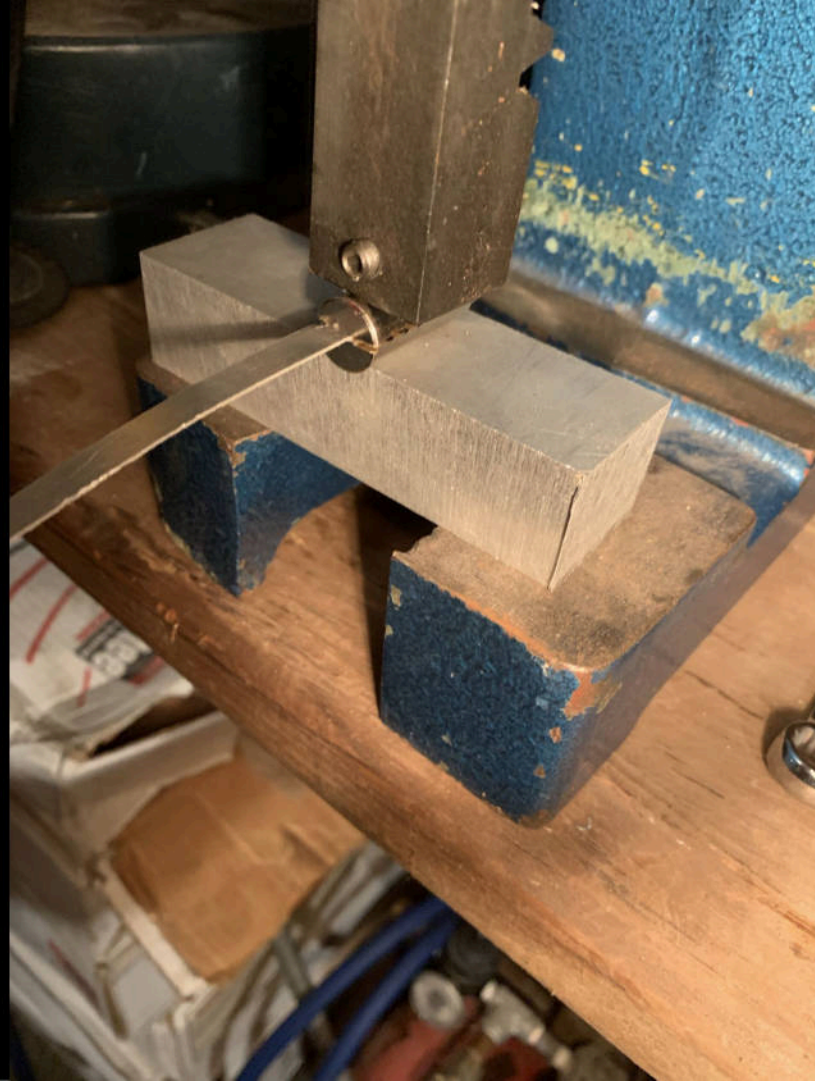














V. 115 SFA  
A 4.25 TIME RATING INTMT. INSUL. CLASS B  
CONTINUOUS DUTY - REOIL EVERY YEAR  
INTERMITTENT DUTY - REOIL EVERY 3 YEARS  
OCCASIONAL DUTY - REOIL EVERY 5 YEARS  
REOIL EACH BEARING WITH 150 DROPS  
(APPROXIMATELY 1 TEASPOON) SAE 30 OIL  
E22922 MPG. NO. M059C ASSEMBLED IN MEXICO  
WARNING: DO NOT OVERTIGHTEN. OVERTIGHTENING WILL BREAK HANDLES AND / OR DAMAGE THREADS.

CAUTION  
OVERTIGHTENING WILL  
BREAK HANDLES AND / OR  
DAMAGE THREADS.

NARROW STAMP

















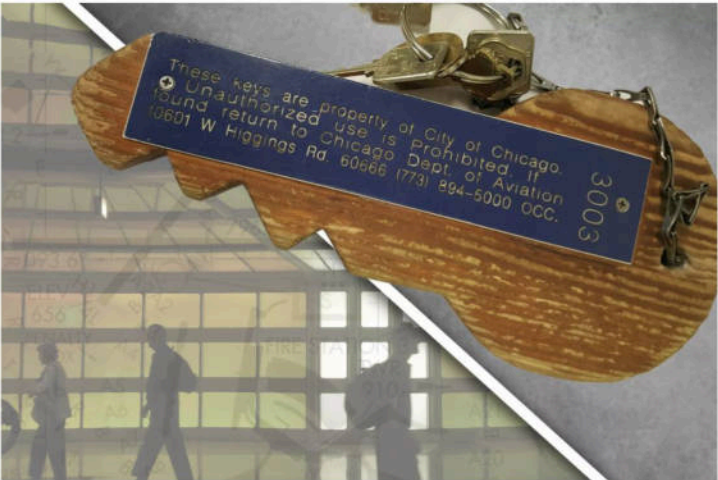
THE WATCHDOGS NEWS CHICAGO

# Master keys for O'Hare Airport security access were lost, costing city in 'five figures'

A set of keys that provides almost total access to O'Hare Airport were lost and never recovered. But there were rare consequences for the employee involved, a Sun-Times investigation found.

By Robert Herguth | Jun 21, 2019, 5:00am CDT

   SHARE



A set of keys like these belonging to the Chicago Department of Aviation and allowing high-level security access to O'Hare Airport were lost. | Photo illustration by Ashlee Rezin & Brian Ernst / Sun-Times

It looks like something you'd borrow from the counter of a gas station to get into the

## MOST READ

School cops speak out at t  
about kids out there that h

Cubs trade pitcher Mike M  
Royals for catcher Martin M

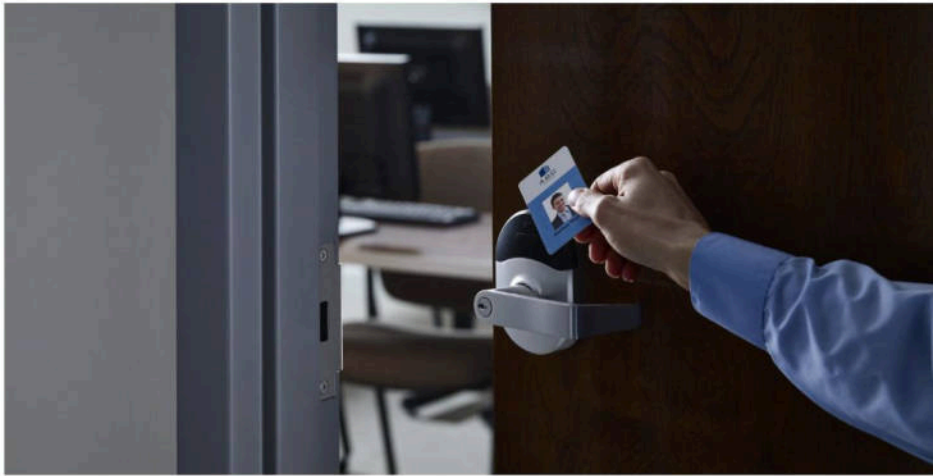
Dach, Nylander, others del  
Blackhawks' development

2 men killed in South Chic  
police

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battle over money

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NEWS

# Lost, stolen campus master keys too expensive to replace

🕒 November 27, 2018 👁 0 view

The master keys at UCA have recently been brought into the light due to an investigation involving a lost key.

The investigation began in June after a theft was reported in Assistant Director of Financial Aid for Scholarships Andrew Linn's office in McCastlain Hall, Jeff Pitchford, vice president of university and government relations, said.

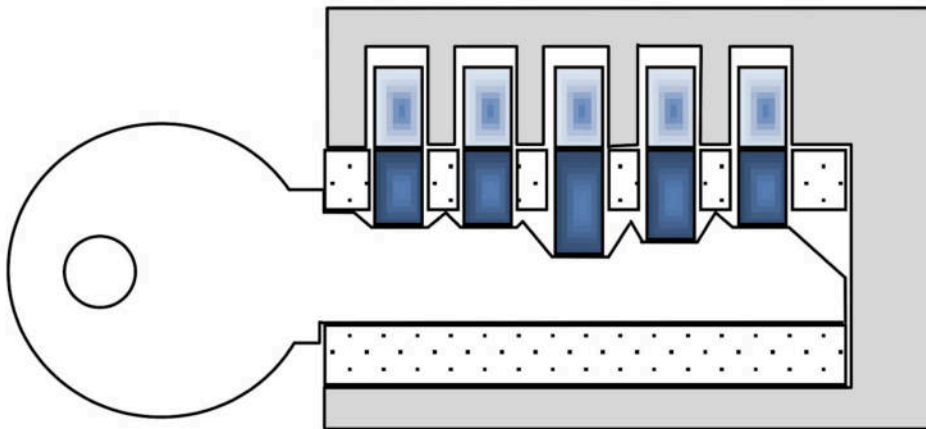
A thief reportedly broke in and stole four pills out of Linn's office. The key used was a grand



# Lost Lock...



+



+



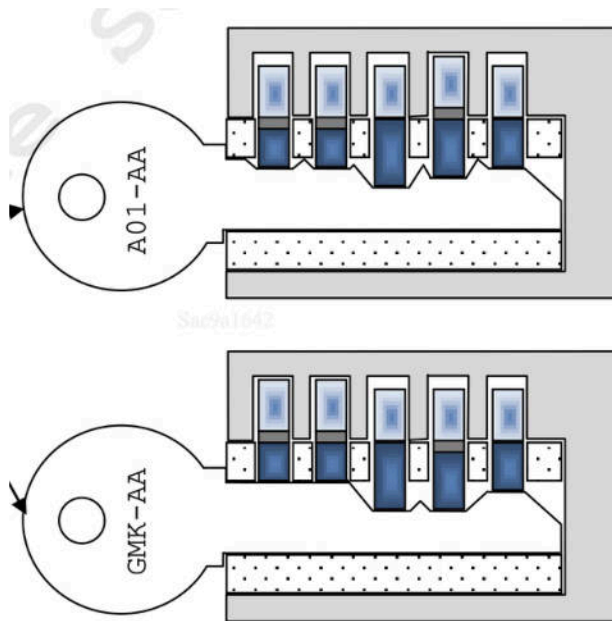
=



# Lost Lock...



+



+



=





Starting space - pins: 6 , depths: 6 , numbered NA . Type of system: Biaxial/m3 Decoding:

Top Level Master

Starting Keyspace

Space: 46656; Angle Space: 46656/11390625

Known shear lines: Pin 1: 1,4 Pin 2: 2,3 Pin 3: 2,5 Pin 4: 3,4 Pin 5: 2,6 Pin 6: 1,3 ☒ Non-MK possible Space: 64 X

MACS of 2

Space: 24 X

IC system in pins 3,4 at depth 3

Space: 4 X

MACS for master control key: 2

Known change key, code 342343 , ☐ No-file ☐ 2-step ☐ Min spacer size

Space: 2 X

Remaining Keyspace:

135421

435421

Space: 2

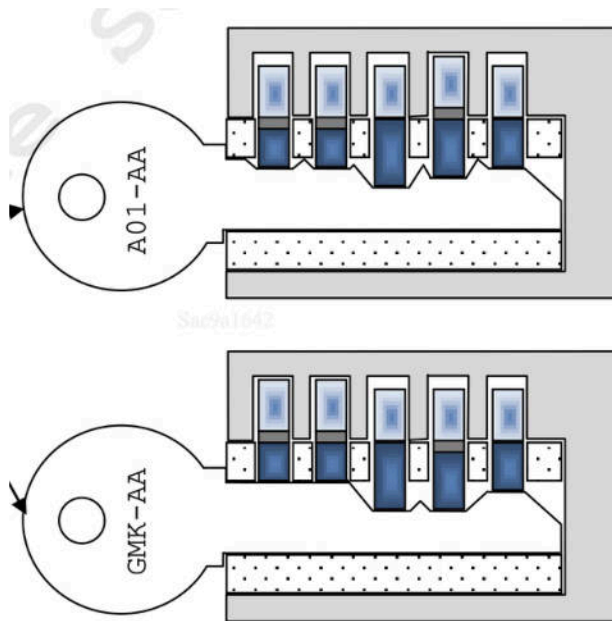
| Depth | Pin 1 | Pin 2 | Pin 3 | Pin 4 | Pin 5 | Pin 6 |
|-------|-------|-------|-------|-------|-------|-------|
|-------|-------|-------|-------|-------|-------|-------|

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 0 | 2 |
| 2 | 0 | 0 | 0 | 0 | 2 | 0 |
| 3 | 0 | 2 | 0 | 0 | 0 | 0 |
| 4 | 1 | 0 | 0 | 2 | 0 | 0 |
| 5 | 0 | 0 | 2 | 0 | 0 | 0 |
| 6 | 0 | 0 | 0 | 0 | 0 | 0 |

# Lost Lock...



+



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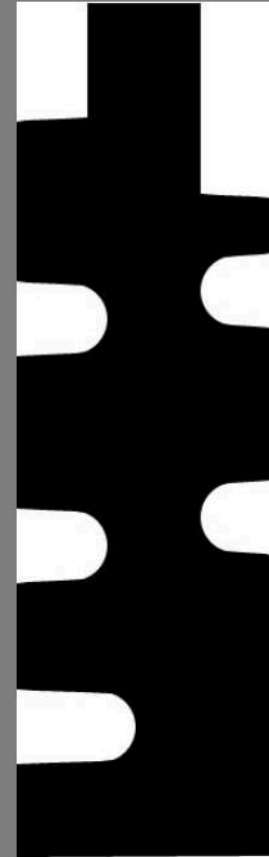
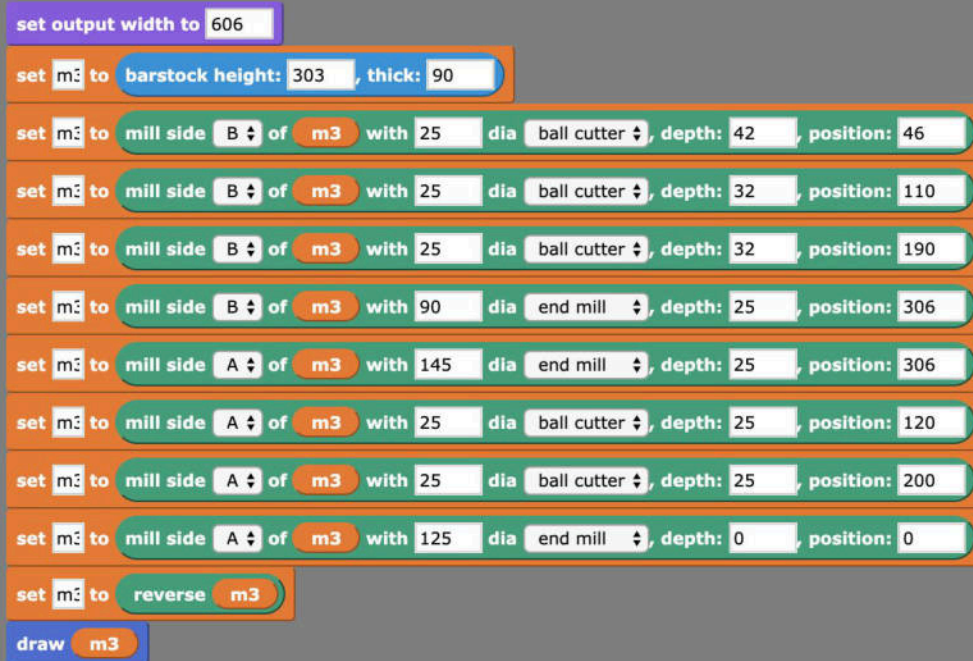




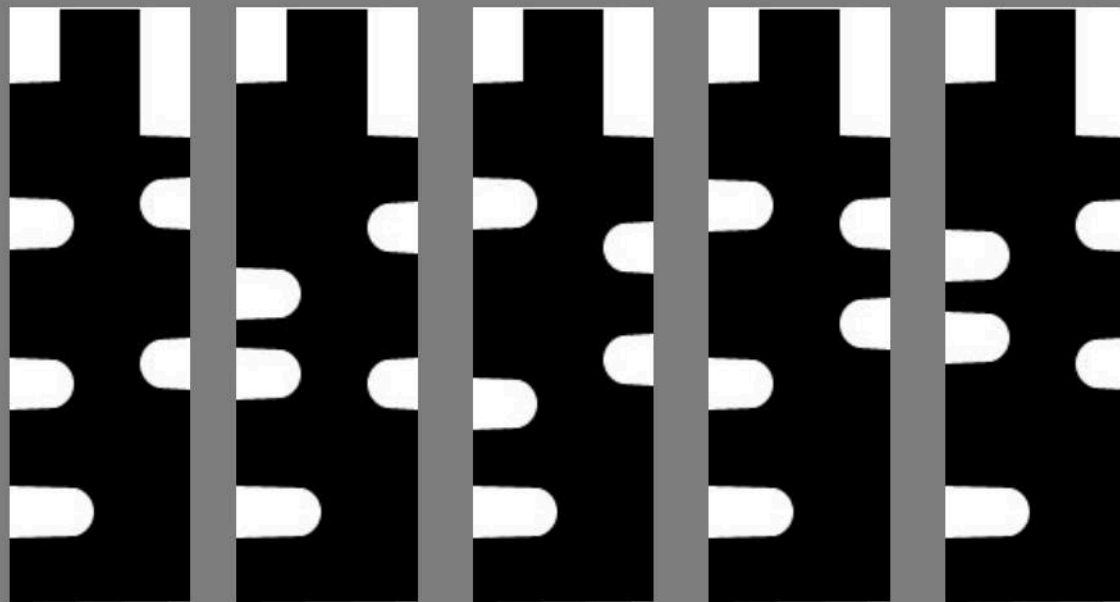




# M3 Blank Manufacturing



$m^3$  Variations





# Master m<sup>3</sup> Keyway

set m3 to barstock height: 298 , thick: 90

set m3 to mill side B of m3 with 25 dia ball cutter , depth: 42 , position: 46

set m3 to mill side B of m3 with 113 dia end mill , depth: 32 , position: 150

set m3 to mill side B of m3 with 90 dia end mill , depth: 25 , position: 306

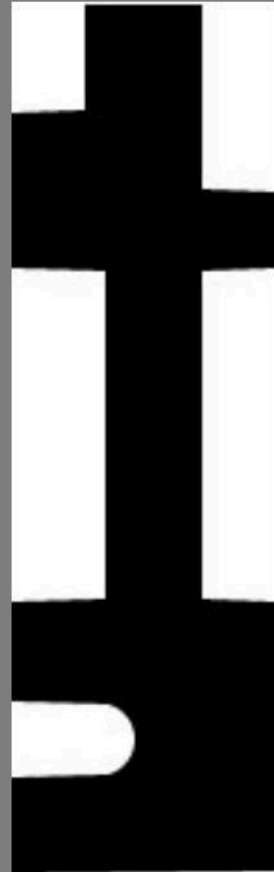
set m3 to mill side A of m3 with 145 dia end mill , depth: 25 , position: 306

set m3 to mill side A of m3 with 113 dia end mill , depth: 25 , position: 150

set m3 to mill side A of m3 with 125 dia end mill , depth: 0 , position: 0

set m3 to reverse m3

draw m3

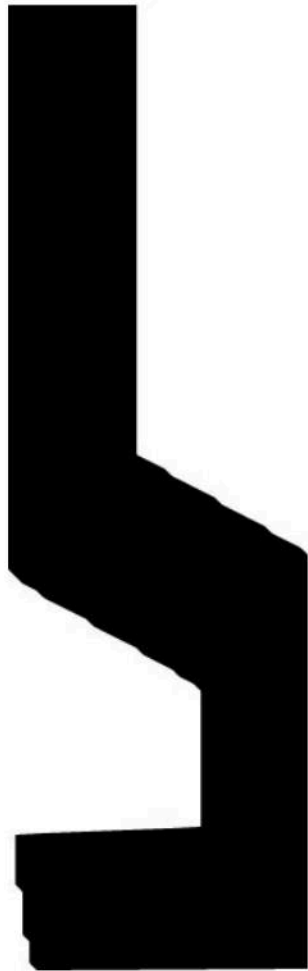


for **i** in all keyways, large DB

write property name of **i**

write metal filed to turn **i** into **m3**

















# Photographing a Lock







compare **Photo**, Medeco 19S 



compare **Photo**, upside-down Medeco 17S 



# Measuring the Key





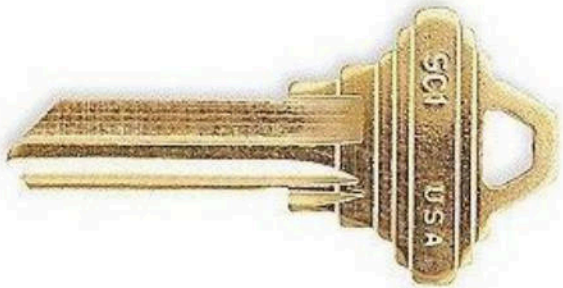
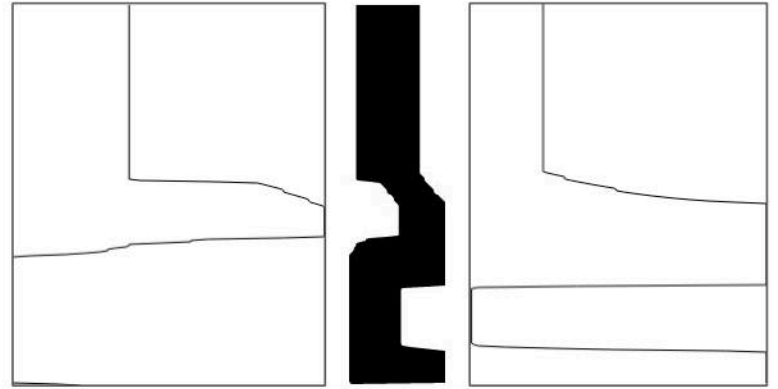
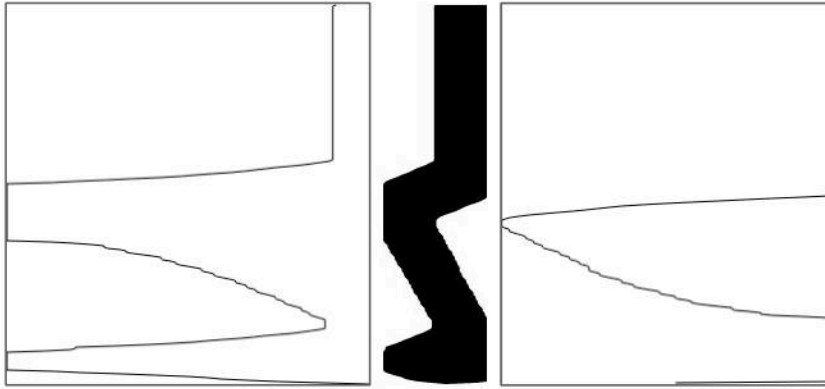


KLOM<sup>ooo</sup>®

KL0M-3100

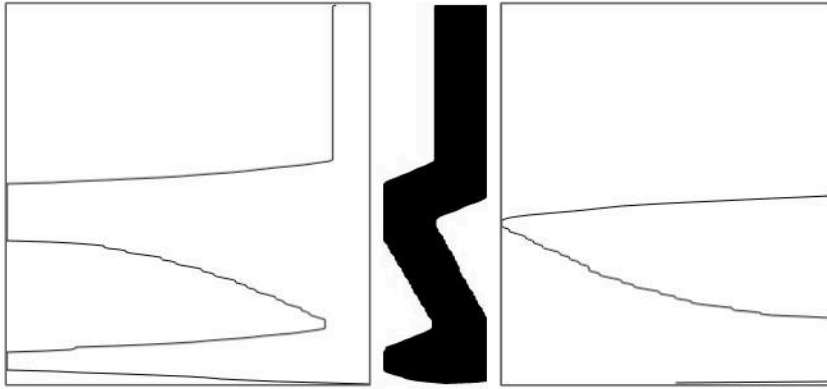
The brand of High Quality

# Photographing the Key



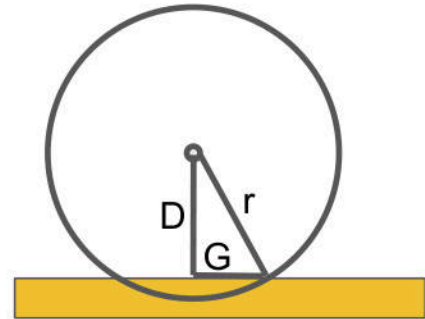
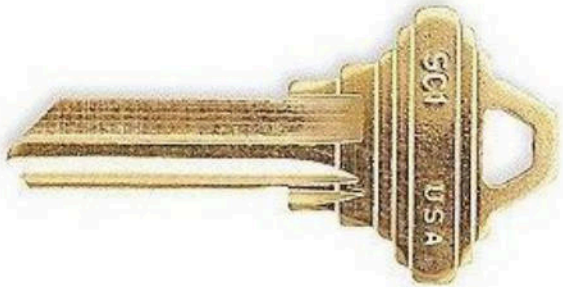


# Milling Marks - Shallow Slope Gives Precision



$$G = \sqrt{r^2 - D^2}$$

$$D = \sqrt{r^2 - G^2}$$







# Outline

## ~~1. Keyways~~

## 2. Other Stuff

- a. Sidewinder Keys
- b. Primus
- c. Assa
- d. Medeco Angled Cuts
- e. Dimple Locks, Mul-T-Lock
- f. Interactive Components, Mul-T-Lock
- g. Disc Detainer Locks, Abloy
- h. Patents
- i. Duplication Authorisation Cards

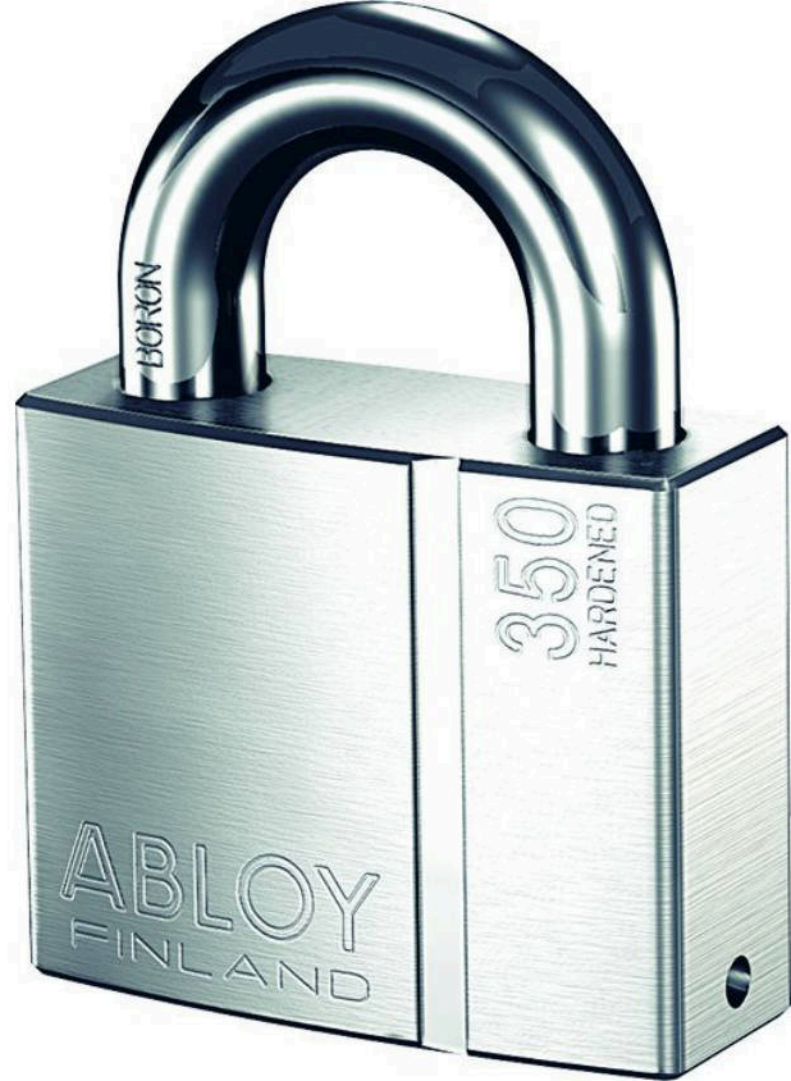




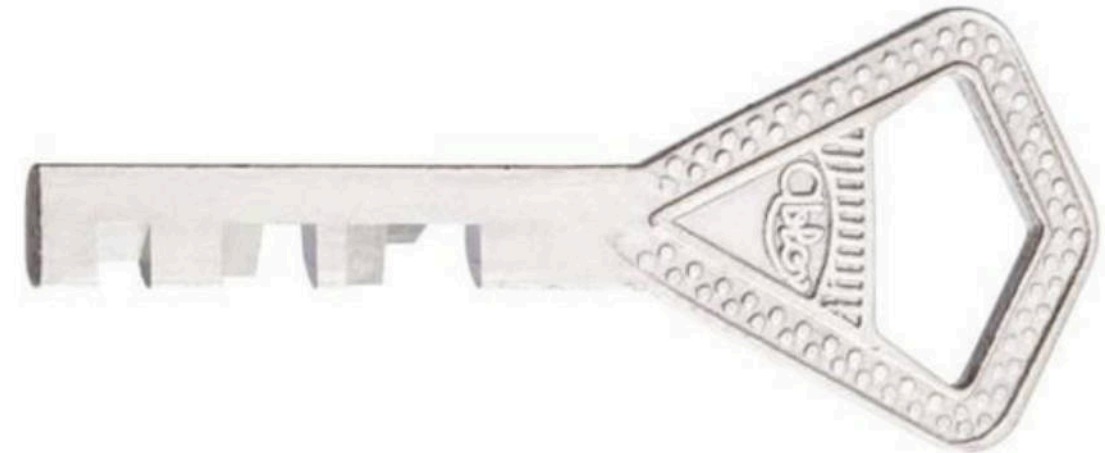


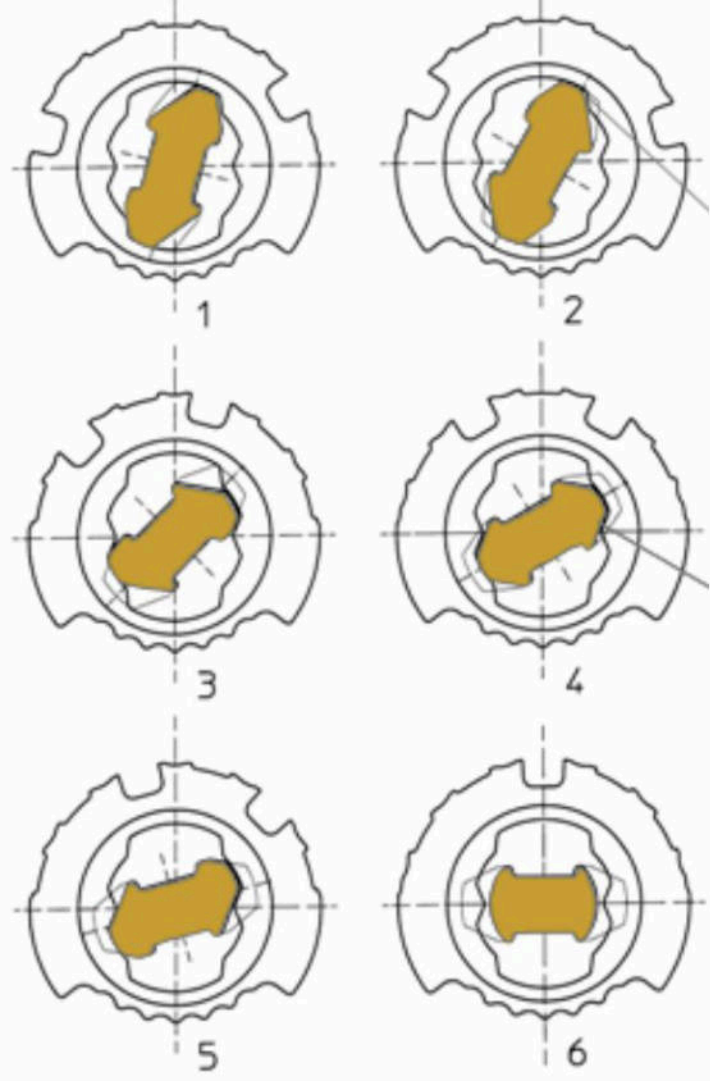




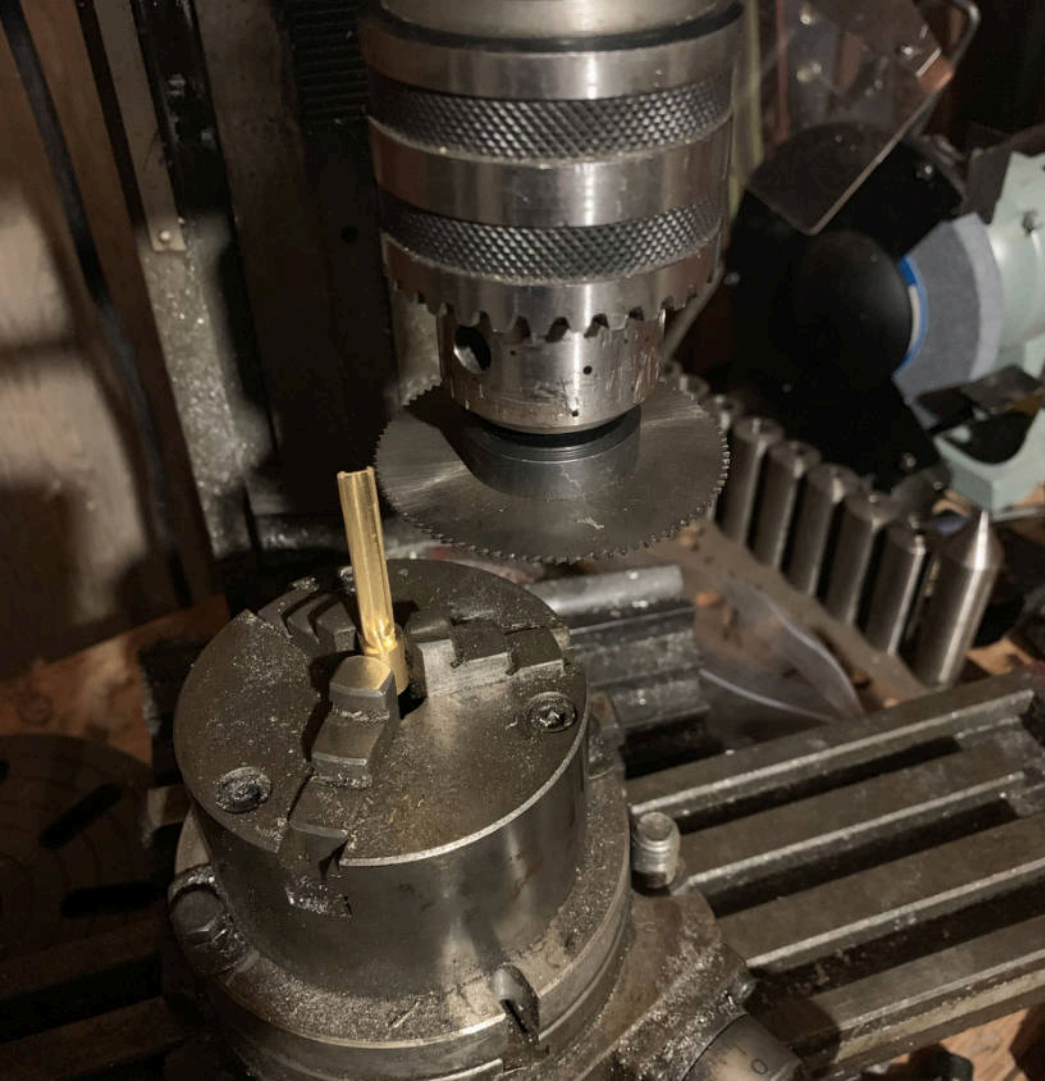




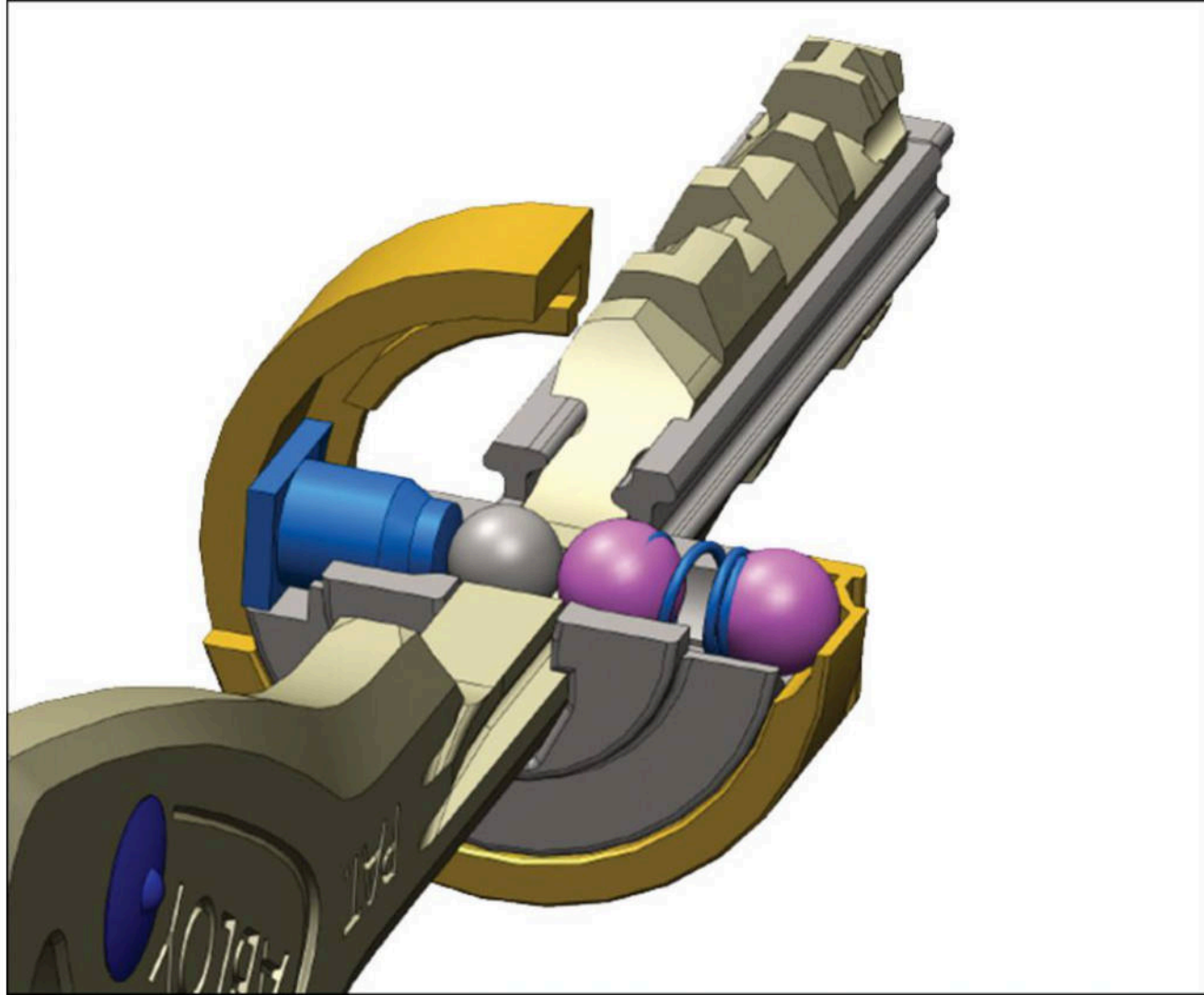


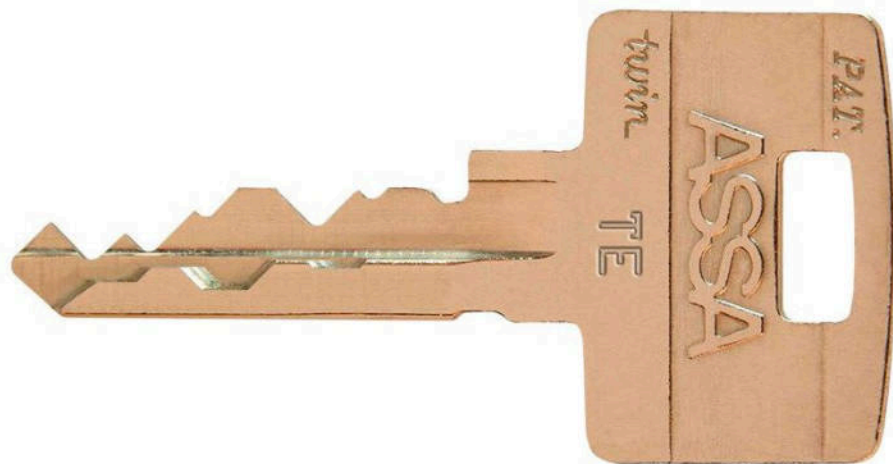


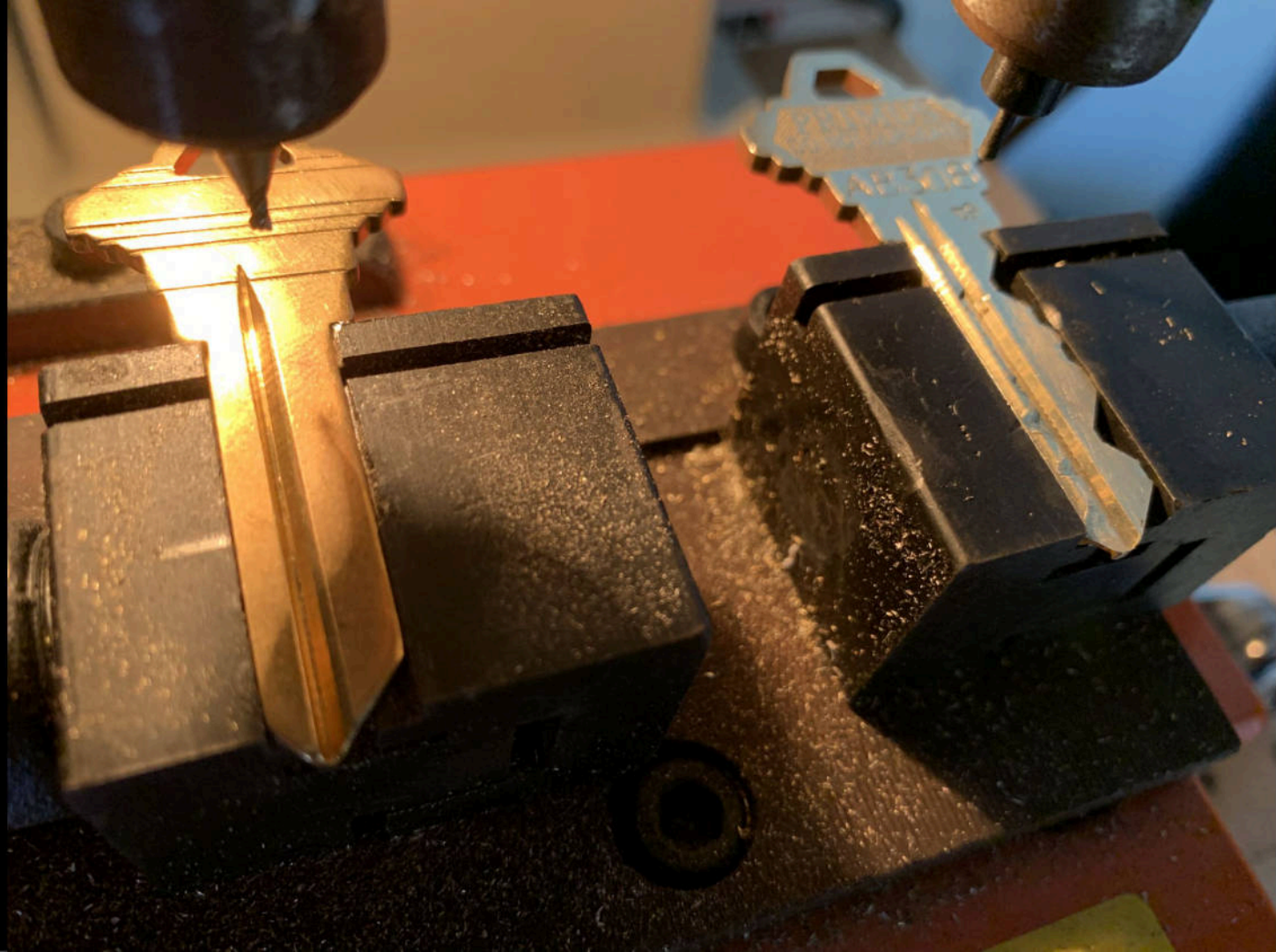




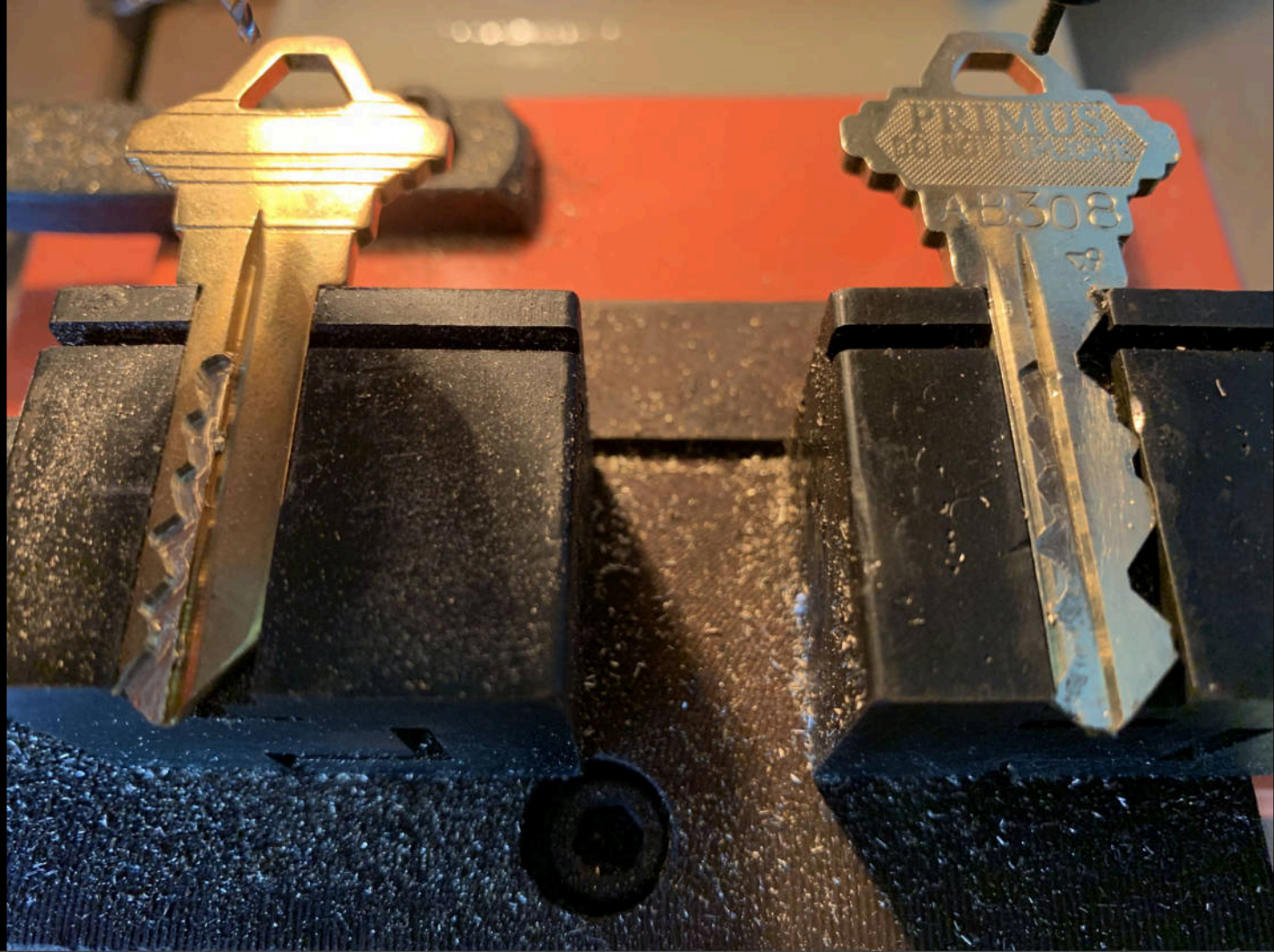




























## MUL-T-LOCK INTERACTIVE 264B KEYWAY PROFILE KEY BLANKS LOCKSMITH SUPPLY 10 pcs

★★★★★ Be the first to [write a review](#).

Condition: **New**

Sale ends in: 06d 02h 26m

Quantity:

2 available / **1 sold in 1 hour**

Was: ~~US \$59.90~~ ?

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Item location: israel, Israel

Ships to: Worldwide [See exclusions](#)

**\$ Have one to sell?**

[Sell now](#)

MT5 Patent = No ebay blanks



Generate code card for angles  and depths   
on keyway





# Outline

- ~~1. Keyways~~
- ~~2. Other Stuff~~
3. Bonus! Remediation

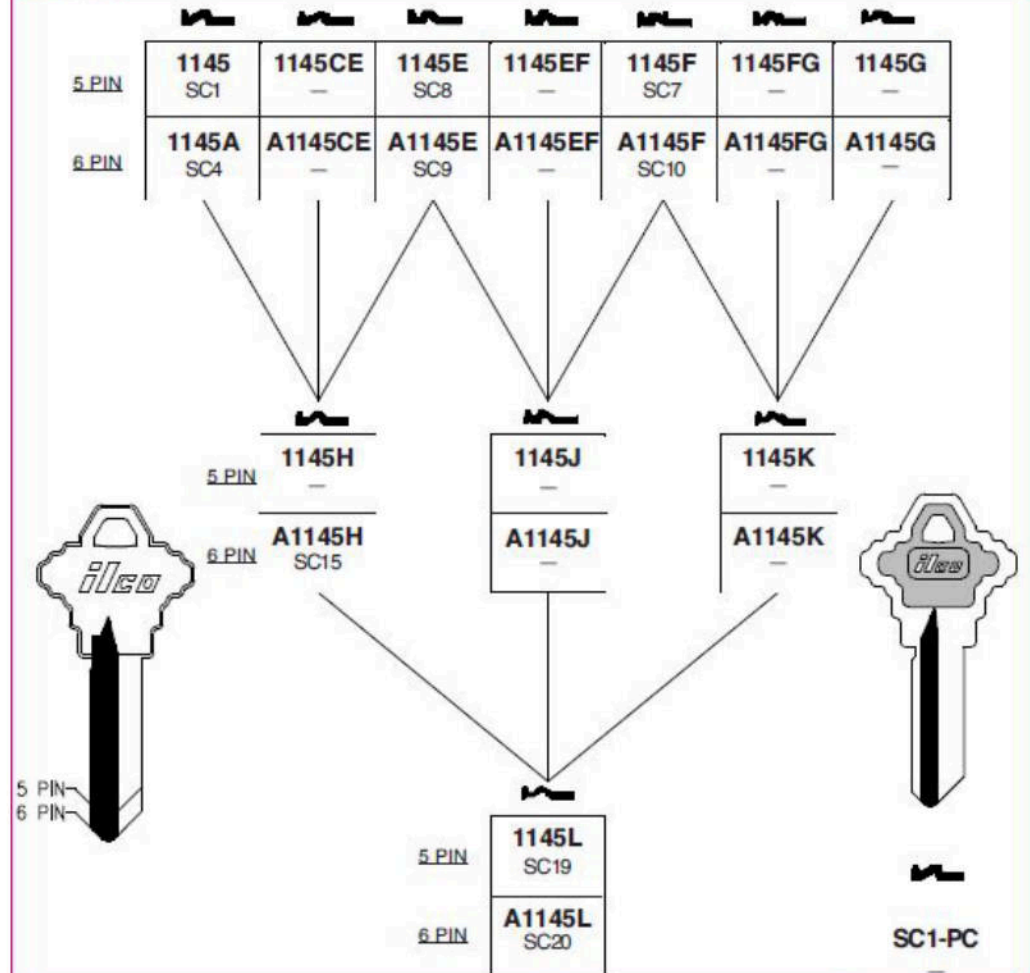
Lost Lock = Lost GMK



=



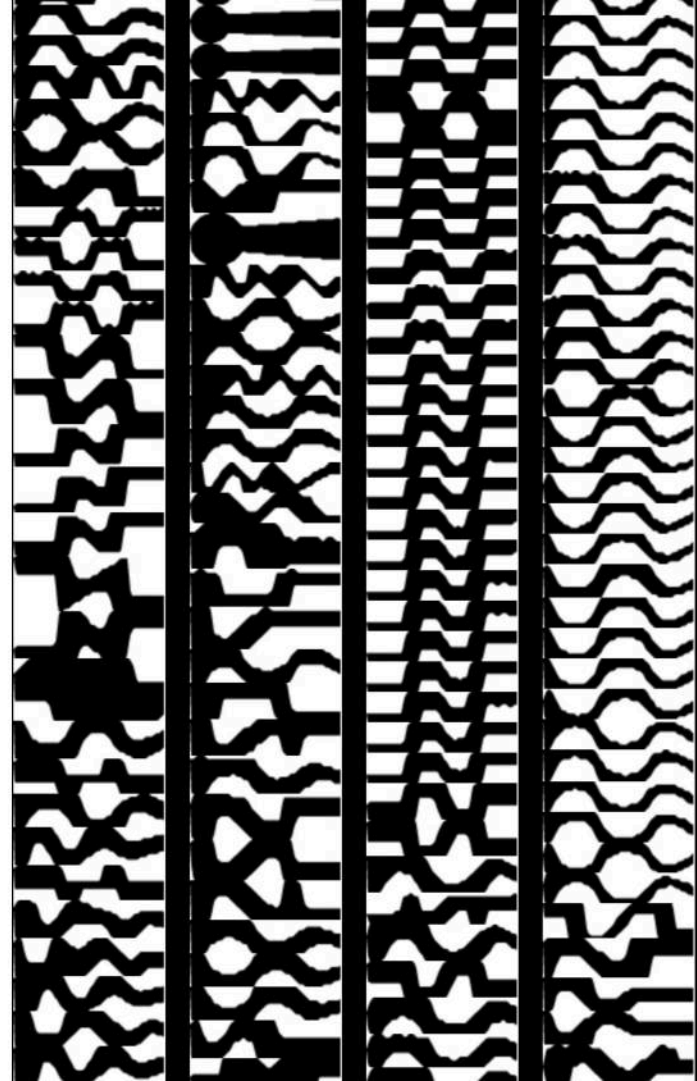
Sectional keyways -  
great as an extra  
security control. Bad as  
the only one.





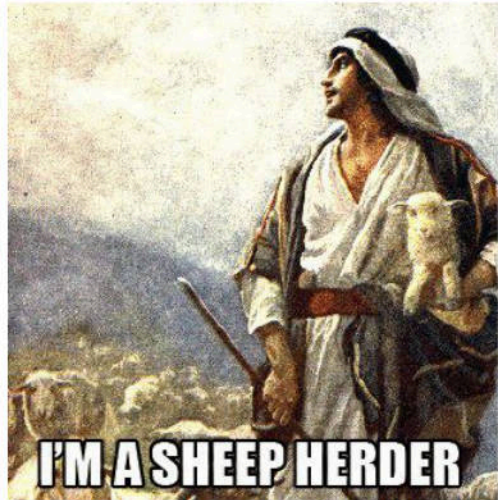
### Restricted keyways -

great as an extra security control. Bad as the only one.



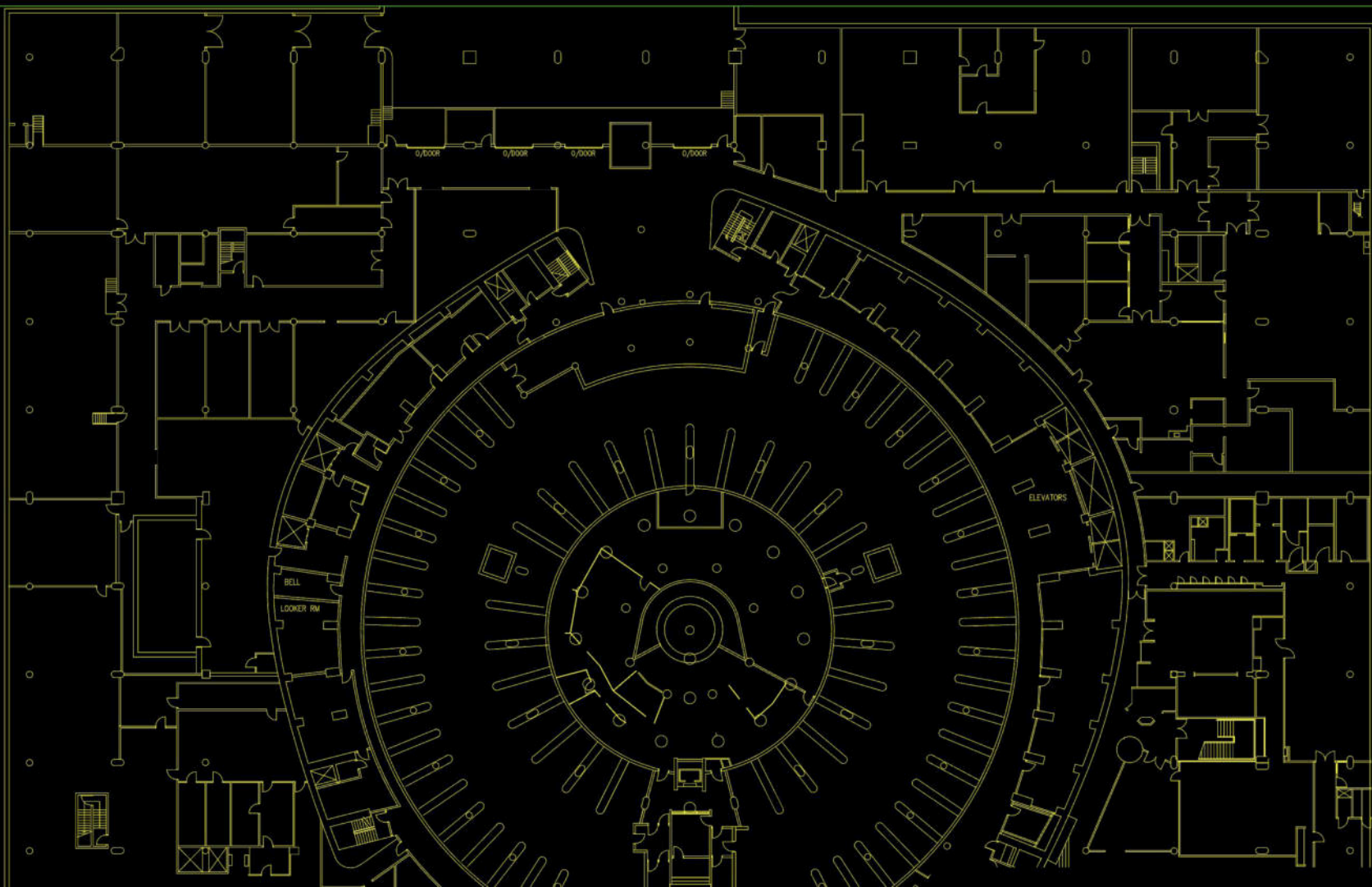
“physical keys are dead hurr durr”

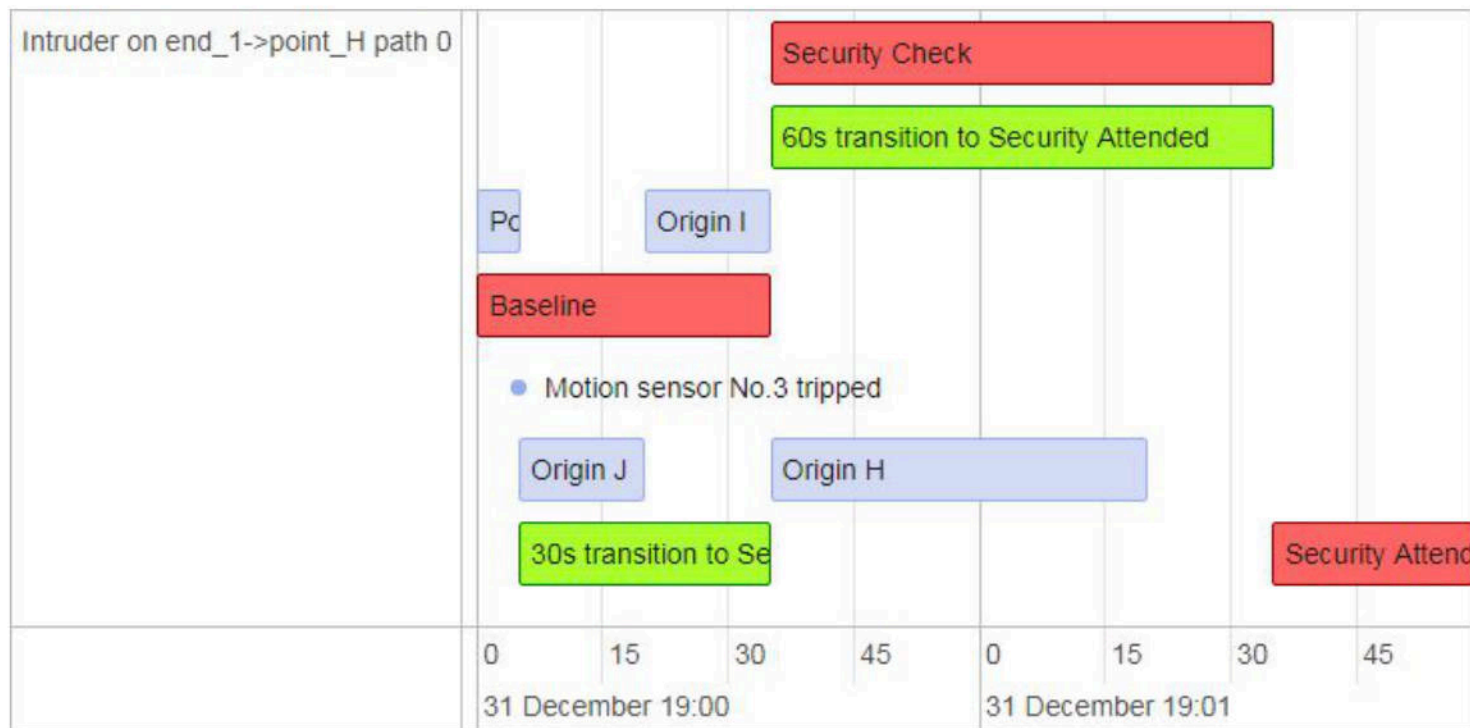
“physical keys are dead hurr durr”





Understand your threat model.







4N6

# Microscopy



# Questions!

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**r.graydon@ggrsecurity.com**